

SIMATS ENGINEERING
Department of Computer Science and Engineering
ASSESSMENT TOOL 2 – COMPARATIVE ANALYSIS

Course Code:	CSA12	Course Title:	Computer Architecture
CO Assessed:	CO4 – Precision (BL4)	Assessment:	Comparative Analysis
Weightage:	30%	Total Marks:	100
CO4 Statement:	Implement hierarchical memory systems by differentiating cache levels (L1, L2, L3), virtual memory with paging, and advanced memory technologies (DRAM, SRAM, NAND Flash, MRAM, RRAM) based on latency, bandwidth, and throughput characteristics.		
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Section / Batch:	BE ECE Final year	Date:	24/08/2026

Instructions to Students

- Identify the relevant concepts of L1, L2, and L3 cache, SRAM, DRAM, NAND Flash, MRAM, RRAM, virtual memory, paging, latency, bandwidth, and throughput.
- Analyze the memory requirements of different computer systems by considering cache levels, memory technologies, access latency, bandwidth, throughput, capacity, and energy consumption.
- Evaluate the performance characteristics of different memory technologies and determine their suitability for cache memory, main memory, secondary storage, and future hierarchical memory systems.
- Compute relevant performance measures such as access time, latency, bandwidth, throughput, hit/miss rate, average memory access time (AMAT), speedup, and energy efficiency wherever applicable.
- Interpret the results by assessing memory bottlenecks, performance improvements, technology trade-offs, and the suitability of each memory technology or memory hierarchy for a given application.

QUESTIONS

1. Analyze the relationship between cache size, cache access time, and cache hit rate. Evaluate how changes in cache size affect processor performance, memory latency, and overall system throughput.
2. Assess the impact of memory bandwidth on data-intensive applications such as artificial intelligence, scientific computing, and multimedia processing. Analyze how insufficient bandwidth can create memory bottlenecks and limit processor performance.
3. Evaluate the effect of page size on virtual memory performance. Analyze how page size influences page faults, memory utilization, translation overhead, and system throughput.
4. Examine the role of data locality in improving cache performance. Analyze how temporal and spatial locality influence cache hit rates, memory access latency, and processor efficiency.
5. Assess the suitability of SRAM, DRAM, NAND Flash, MRAM, and RRAM for different levels of a memory hierarchy. Evaluate each technology based on latency, bandwidth, capacity, energy consumption, endurance, and cost, and recommend appropriate technologies for specific applications.

Code of Conduct:

I Deepan Kumar P(192312498) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

RUBRICS FOR CALCULATION

Criteria	Weightage(%)	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Scope of Items	20%	Selects highly relevant items with clear scope and justification	Selects appropriate items with minor gaps	Selection is limited or partially relevant	Items are unclear or not relevant
Comparison Depth	25%	Provides detailed and comprehensive comparison across multiple aspects	Provides adequate comparison with some depth	Limited comparison with few aspects	Very minimal or no comparison
Use of Evidence	20%	Supports comparison with accurate data, examples, or references	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Critical Thinking	20%	Demonstrates strong critical thinking with clear interpretation and reasoning	Shows reasonable interpretation with minor gaps	Basic interpretation; lacks depth	No meaningful interpretation
Clarity of Presentation	15%	Well-organized, clear, and logically structured presentation	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

Marks Evaluation:

Q. No	Scope of Items (20)	Comparison Depth (25)	Use of Evidence (20)	Critical Thinking (20)	Clarity of Presentation (15)	Total (out of 100)
1	/ 4	/ 4	/ 4	/ 4	/ 4	
2	/ 4	/ 4	/ 4	/ 4	/ 4	
3	/ 4	/ 4	/ 4	/ 4	/ 4	
4	/ 4	/ 4	/ 4	/ 4	/ 4	
5	/ 4	/ 4	/ 4	/ 4	/ 4	
	/ 20	/ 25	/ 20	/ 20	/ 15	/ 100

Course Coordinator

Faculty Incharge

Q1. Relationship Between Cache Size, Cache Access Time and Cache Hit Rate.

Cache memory is a small, high-speed memory located close to the CPU. It stores frequently used data and instructions to reduce the time required to access main memory.

When cache size increases, more data can be stored, so the cache hit rate generally increases. A higher hit rate means fewer cache misses and fewer accesses to slower main memory. This improves processor performance and reduces memory latency.

However, a larger cache may have a slightly higher access time, along with increased cost and power consumption. Therefore, cache size must be balanced with speed.

The performance can be measured using:

$$\text{AMAT} = \text{Hit Time} + (\text{Miss Rate} \times \text{Miss Penalty})$$

For example, if:

- Hit time = 2 ns
- Hit rate = 95%
- Miss penalty = 50 ns

Then:

$$\text{AMAT} = 2 + (0.05 \times 50) = 4.5 \text{ ns}$$

If the cache size is increased and the hit rate becomes 98%:

$$\text{AMAT} = 2 + (0.02 \times 50) = 3 \text{ ns}$$

Thus, the higher hit rate reduces average memory access time.

Cache Size	Hit Rate	Access Time	Performance
Small	Lower	Faster	Lower
Medium	Higher	Moderate	Better
Large	Higher	May increase	Generally better

Therefore, increasing cache size generally improves hit rate, memory latency, processor performance, and system throughput. However, an excessively large cache can increase access time, cost, and power consumption. Hence, an optimal cache size provides the best balance between speed and capacity.

Q2. Impact of Memory Bandwidth on Data-Intensive Applications

Memory bandwidth is the amount of data that can be transferred between the processor and memory in a given amount of time, usually measured in **GB/s**. It is very important for data-intensive applications such as **artificial intelligence, scientific computing, and multimedia processing**.

In **artificial intelligence**, large amounts of data, model parameters, and intermediate results must be continuously transferred between memory and the processor. Higher memory bandwidth allows AI systems to process data faster.

In **scientific computing**, applications such as simulations and numerical calculations frequently access large datasets. Insufficient bandwidth can make the processor wait for data.

In **multimedia processing**, image, audio, and video applications require continuous transfer of large amounts of data. Higher bandwidth helps maintain smooth and fast processing.

Effect of Insufficient Bandwidth

When memory bandwidth is lower than the processor's data requirement, a **memory bottleneck** occurs. The processor has to wait for data instead of performing computations.

For example, if a processor requires **100 GB/s** but the memory provides only **50 GB/s**, the processor cannot receive data at its full processing capacity.

Therefore:

Higher bandwidth → Faster data transfer → Better processor utilization → Higher throughput

Lower bandwidth → Data waiting → Memory bottleneck → Lower processor performance

Conclusion

Memory bandwidth directly affects the performance of data-intensive applications. AI, scientific computing, and multimedia applications require high bandwidth because they process large amounts of data. Therefore, increasing memory bandwidth can reduce memory bottlenecks, improve processor utilization, reduce execution time, and increase overall system throughput.

Q3. Effect of Page Size on Virtual Memory Performance

Virtual memory divides memory into fixed-size blocks called **pages**. The size of these pages affects page faults, memory utilization, translation overhead, and overall system throughput.

A **small page size** requires more pages to represent the same amount of memory. This can increase the number of page-table entries and translation overhead. However, it reduces memory wastage because smaller portions of memory are allocated.

A **large page size** requires fewer pages and therefore fewer page-table entries. It can also reduce translation overhead and improve performance when programs access nearby data. However, it may cause more memory wastage if the entire page is not used.

Factor	Small Page Size	Large Page Size
Page faults	May be more frequent	May be fewer
Memory utilization	Better	More wastage possible
Page-table entries	More	Fewer
Translation overhead	Higher	Lower
Data transferred per page	Less	More
Performance	Suitable for small/random access	Suitable for sequential/local access

For example, for **1 MB of memory**:

4 KB pages:

$1 \text{ MB} \div 4 \text{ KB} = \mathbf{256 \text{ pages}}$

16 KB pages:

$1 \text{ MB} \div 16 \text{ KB} = \mathbf{64 \text{ pages}}$

Thus, larger pages require fewer page-table entries and can reduce translation overhead.

Conclusion

Small pages provide better memory utilization, while large pages reduce page-table and translation overhead. Therefore, the appropriate page size depends on the application's memory access pattern. A suitable page size can reduce overhead, improve memory performance, and increase system throughput.

Q4. Role of Data Locality in Improving Cache Performance

Data locality means that programs tend to access the same data or nearby data repeatedly. It improves cache performance by increasing the probability of finding required data in the cache.

There are two main types of locality:

1. Temporal Locality

Temporal locality means that **recently accessed data is likely to be accessed again soon**.

For example, if a loop repeatedly uses the same variable, keeping that variable in the cache avoids repeated accesses to main memory.

Effect:

Higher cache hit rate → Lower memory latency → Better processor efficiency.

2. Spatial Locality

Spatial locality means that **when one memory location is accessed, nearby memory locations are likely to be accessed soon**.

For example, when accessing elements of an array sequentially, nearby elements are likely to already be available in the same cache block.

Effect:

Better cache utilization → Fewer cache misses → Lower memory access time.

Factor	Temporal Locality	Spatial Locality
Main idea	Reuse the same data	Access nearby data
Example	Repeated variable access	Sequential array access
Cache effect	Increases hit rate	Increases hit rate
Latency	Reduced	Reduced
Processor efficiency	Improved	Improved

Therefore, both temporal and spatial locality improve **cache hit rate**, **reduce memory access latency**, and **increase processor efficiency**. Good data locality allows the CPU to spend less time waiting for data from slower memory.

Q5. Suitability of SRAM, DRAM, NAND Flash, MRAM and RRAM

Different memory technologies have different characteristics in terms of **latency, bandwidth, capacity, energy consumption, endurance, and cost**. Therefore, each technology is suitable for a different level of the memory hierarchy.

Memory	Latency	Capacity	Endurance	Energy	Cost	Suitable Application
SRAM	Very Low	Low	Very High	Higher	High	CPU Cache
DRAM	Low	High	Very High	Moderate	Moderate	Main Memory
NAND Flash	Higher	Very High	Limited writes	Low	Low	SSD/Storage
MRAM	Low	Medium	Very High	Low	Higher	Non-volatile Memory
RRAM	Low	High potential	Good	Low	Developing	Future Memory

SRAM

SRAM is very fast and has very low latency. It is expensive and has lower storage density.

Best suited for L1, L2 and L3 cache memory.

DRAM

DRAM provides higher capacity at a lower cost than SRAM. It is slower than SRAM and requires periodic refreshing.

Best suited for main memory.

NAND Flash

NAND Flash is non-volatile and provides high storage capacity at low cost. However, it has higher latency and limited write endurance.

Best suited for SSDs and secondary storage.

MRAM

MRAM is a non-volatile memory with low latency and high endurance. It retains data even when power is removed.

Best suited for fast non-volatile and embedded memory applications.

RRAM

RRAM is a non-volatile technology with potential for high density and low energy consumption. It is suitable for emerging and future memory systems.

Best suited for future high-density and low-power applications.

Conclusion:

SRAM → Cache

DRAM → Main Memory

NAND Flash → Secondary Storage

MRAM → Fast Non-volatile Memory

RRAM → Future Memory Applications

Therefore, a combination of these memory technologies provides an effective memory hierarchy by balancing **speed, capacity, energy consumption, endurance, and cost**.